KAIZEN MATHS

Lesson starter

Trig Identities, Proofs and Equations

Proofs & Hence Solve

Name _____

S1

Prove that $\dfrac{1}{1+\tan^2 x} \equiv \cos^2 x.$

S2

Prove that $\dfrac{\sec^2 x - 1}{\sec^2 x} \equiv \sin^2 x.$

S3

Prove that $\dfrac{1-\sin x}{\cos x} \equiv \sec x - \tan x.$

S4

Prove that $\dfrac{1+\tan^2 x}{1+\cot^2 x} \equiv \tan^2 x.$

Show your working

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Exit ticket

Trig Identities, Proofs and Equations

Proofs & Hence Solve

Name _____

EA1

(a) Prove that $\cos\theta + \sin\theta\tan\theta \equiv \sec\theta$.(b) Hence find the exact roots of $\cos\theta + \sin\theta\tan\theta = 4\cos\theta$, for $0 \leq \theta \leq 2\pi$.

EA2

(a) Prove that $\dfrac{1+\tan^2 x}{1+\cot^2 x} \equiv \tan^2 x$.(b) Hence solve, for $0 \leq x \leq 360°$, $\dfrac{1+\tan^2 x}{1+\cot^2 x} = 3$.

Show your working

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Exit ticket

Trig Identities, Proofs and Equations

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EB1

Prove that $\text{cosec}^2\theta + 4\,\text{cosec}\,\theta + 8 = 0$ has no real solutions.

Show your working

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Exit ticket

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(a) Prove that $\cos\theta + \sin\theta\tan\theta \equiv \sec\theta$.(b) Hence find the exact roots of $\cos\theta + \sin\theta\tan\theta = 4\cos\theta$, for $0 \leq \theta \leq 2\pi$.

EA2

(a) Prove that $\dfrac{1+\tan^2 x}{1+\cot^2 x} \equiv \tan^2 x$.(b) Hence solve, for $0 \leq x \leq 360°$, $\dfrac{1+\tan^2 x}{1+\cot^2 x} = 3$.

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Teacher answers & working

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S1

Prove that $\dfrac{1}{1+\tan^2 x} \equiv \cos^2 x$.

Answer:

Identity proven

Working:

Use $1+\tan^2 x = \sec^2 x$.

$\dfrac{1}{1+\tan^2 x} = \dfrac{1}{\sec^2 x}$.

Since $\sec x = \dfrac{1}{\cos x}$, $\dfrac{1}{\sec^2 x} = \cos^2 x$.

Identity proven

S2

Prove that $\dfrac{\sec^2 x - 1}{\sec^2 x} \equiv \sin^2 x$.

Answer:

Identity proven

Working:

Use $\sec^2 x - 1 = \tan^2 x$.

$\dfrac{\sec^2 x - 1}{\sec^2 x} = \dfrac{\tan^2 x}{\sec^2 x}$.

$= \dfrac{\sin^2 x / \cos^2 x}{1/\cos^2 x}$.

$= \sin^2 x$.

Identity proven

S3

Prove that $\dfrac{1-\sin x}{\cos x} \equiv \sec x - \tan x$.

Answer:

Identity proven

Working:

Start with the right hand side.

$\sec x - \tan x = \dfrac{1}{\cos x} - \dfrac{\sin x}{\cos x}$.

$= \dfrac{1-\sin x}{\cos x}$.

This matches the left hand side.

Identity proven

Teacher answers & working

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S4

Prove that $\dfrac{1+\tan^2 x}{1+\cot^2 x} \equiv \tan^2 x$.

Answer:

Identity proven

Working:

Use $1+\tan^2 x = \sec^2 x$ and $1+\cot^2 x = \operatorname{cosec}^2 x$.

$$\frac{1+\tan^2 x}{1+\cot^2 x} = \frac{\sec^2 x}{\operatorname{cosec}^2 x}.$$

$$= \frac{1/\cos^2 x}{1/\sin^2 x}.$$

$$= \frac{\sin^2 x}{\cos^2 x} = \tan^2 x.$$

Identity proven

EA1

(a) Prove that $\cos\theta + \sin\theta\tan\theta \equiv \sec\theta$.(b) Hence find the exact roots of $\cos\theta + \sin\theta\tan\theta = 4\cos\theta$, for $0 \leq \theta \leq 2\pi$.

Answer:

$\theta = \dfrac{\pi}{3}, \dfrac{2\pi}{3}, \dfrac{4\pi}{3}, \dfrac{5\pi}{3}$

Working:

From the proof, replace $\cos\theta + \sin\theta\tan\theta$ with $\sec\theta$.

$$\sec\theta = 4\cos\theta$$

$$\frac{1}{\cos\theta} = 4\cos\theta$$

$$1 = 4\cos^2\theta$$

$$\cos^2\theta = \frac{1}{4}$$

$$\cos\theta = \pm\frac{1}{2}$$

$$\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

Teacher answers & working

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EA2

(a) Prove that $\dfrac{1+\tan^2 x}{1+\cot^2 x} \equiv \tan^2 x$.(b) Hence solve, for $0 \leq x \leq 360°$, $\dfrac{1+\tan^2 x}{1+\cot^2 x} = 3$.

Answer:

$x = 60°, 120°, 240°, 300°$

Working:

Replace the numerator and denominator using identities.

$$\frac{1+\tan^2 x}{1+\cot^2 x} = \frac{\sec^2 x}{\operatorname{cosec}^2 x}$$

$$= \frac{1/\cos^2 x}{1/\sin^2 x} = \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$$

Use the identity in the equation.

$\tan^2 x = 3$

$\tan x = \pm\sqrt{3}$

$x = 60°, 120°, 240°, 300°$

EB1

Prove that $\operatorname{cosec}^2\theta + 4\operatorname{cosec}\theta + 8 = 0$ has no real solutions.

Answer:

No real solutions

Working:

Let $u = \operatorname{cosec}\theta$.

$u^2 + 4u + 8 = 0$

Calculate the discriminant.

$b^2 - 4ac = 4^2 - 4(1)(8) = 16 - 32 = -16$

A negative discriminant means there are no real values of u.

Therefore there are no real values of θ that satisfy the original equation.